

CASINO BUSINESS PRESENTATION

SPIT SHINE

The Variable-Table Game of Show, Spit, Shine & Win

Complete Deal & Play Procedures, Rules, Odds, Audit Data, and Integrity Controls



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1. Executive Summary

Spit Shine is a proprietary casino table game for one to four players against the dealer, played with a 104-card shoe (two standard 52-card decks). Each seat receives four cards and, across seven clearly defined phases - Private Reveal, Showing, Bow Out, Spit, Shine, Selection, and Showdown - builds the highest-scoring two-card hand. The game combines the bluffing depth of poker, the fixed-dealer structure of blackjack, and the clean point-comparison of baccarat, but its signature mechanics (Spit card exchange and Shine redraw) create a level of player agency absent from most table games.

The table is not fixed at any player count. It seats any number of players from one to four against the dealer, and players may join or drop out freely between rounds: a player who joins the next round posts an ante and receives a fresh hand, a player who leaves simply does not post for the next round, and empty seats post no ante and take no cards. The pot, rake, and payouts scale with the occupied seats, and every seat chooses its own ante within the published 1-500 chip range.

This guide is the complete operating manual for the game: how to deal it, how to play it, every rule and resolution, the full statistical audit from every perspective (player, dealer, house, regulator), and the integrity controls required for a regulated casino environment. Every figure in this document is reproducible from the canonical engine and the audited simulation dataset.

At a glance

Attribute	Value
Game	Point-comparison table game with seven decision phases
Players	1-4 player seats vs the dealer (heads-up to full table); players drop in/out between rounds
Equipment	2 standard 52-card decks (104 cards), fresh shoe per round, cut card recommended
Ante	Variable per seat, 1-500 chips (denominations 1, 5, 25, 100, 500)
House edge	25% pot rake in all modes; audited hold 24.5% (competitive)
Modes	Competitive (dealer plays) and Cooperative (players vs dealer - flagged for rebalance)
Round duration	2-3.5 minutes by pace (17-30 rounds/hour)
Reference	Canonical engine v11; audit SS-AUDIT-2026-001; 20,000-round seeded Monte Carlo per scenario

Headline audited numbers

Scenario	Key figure	Detail
Full table (4 players + dealer, 100 ante)	Dealer wins 24.6%; YOU 17.8%	YOU EV -33.9/round; house 122.5/round
Solo heads-up (competitive, 100 ante)	YOU wins 48.0%	YOU EV -26.8/round; house 49.4/round (24.7% hold)
Variable table (drop-in/out, mixed 1-500 antes)	Average 3.59 players	Handle 565 chips/round; house 138.4/round; hold 24.48%
Cooperative full table	Team wins 64.2% of rounds	Player EV -46.1/round; rebalance package in Section 14.5
Double Aces	2.90% per hand; 13.92% dealt per round	A+A = 1000 points; ~17% of rounds finish with A+A after Spit/Shine

Scenario	Key figure	Detail
Exact-tie rounds	3.2% of rounds	Pot splits among tied winners; rounding rake to house
One-million-round audit	Competitive: YOU 17.73%, dealer 24.66%	Section 14.10 - 3 x 1M rounds with 95% CI and charts
Two-million-round hand audit	10M hands tracked; hold 24.51%; 24h model \$70.6k/day house win	Section 14.11 - per-round hand/point tracking + 24h charts

Ambiguities resolved in this edition

- Exact ties split the pot (score -> rank -> suit -> split); any integer remainder is retained by the house as rounding rake and logged.
- The Spit-received card is private to the spitter; a target can receive at most one Spit per round; the dealer never Spits or Bows Out.
- Tables are variable: 1-4 player seats, per-seat antes from 1-500 chips, players drop in/out between rounds, empty seats post no ante.
- Mixed antes transfer EV toward small bettors, so a 5x ante spread limit is recommended (Section 14.9).
- Cooperative mode's current economics are house-favorable; a rebalance package (dealer 1-card Shine, dealer ante, 125-chip team payout) brings player EV to about -1.8 chips/round.
- Every remaining edge case and official ruling is collected in Section 15.

How to use this guide

Reader	Start here
Dealers / table supervisors	Sections 7-10: procedures, phase-by-phase duties, payout math, rulings
Players	Section 8 and Appendix B: phase guide and a complete worked round
Operators / business development	Sections 3, 11-13: modes, limits, turnover model, canonical rules
Regulators / certification labs	Sections 14-16: audit data, integrity controls, edge-case rulings

Revision history

Version	Date	Changes
1.0	August 6, 2026	Initial 24-page presentation guide
1.1	August 6, 2026	Audit finalized (SS-AUDIT-2026-001); companion references
1.2	August 6, 2026	Variable tables: 1-4 players, drop-in/out, 1-500 chip antes, heads-up solo (36 pages)
1.3	August 6, 2026	Expanded executive summary, edge-case rulings (Section 16), decision flowcharts, phase-state depictions, and additional charts and examples
1.4	August 6, 2026	Dedicated roles section (Player, Dealer, Pit Manager) with integrity watchlists; executive summary states the flexible 1-4 player table and drop-in/out rules explicitly
1.5	August 6, 2026	One-million-round verification: 3 x 1,000,000 rounds (competitive, variable table, solo) with 95% confidence intervals and new metric charts (Section 14.10)

Version	Date	Changes
1.6	August 6, 2026	Two-million-round hand audit: every seat's hand and point totals tracked (10M hands), aggregates and a 5,000-round raw sample, plus a 24-hour distribution model (Section 14.11)

The audit's headline finding is that the game's published odds were materially incorrect (the Double-Aces frequency was understated by roughly 30x and the house edge by 5x). This guide presents the corrected, simulation-verified numbers and the canonical rule set that every future implementation must follow.

2. Game Overview & Design Intent

Spit Shine is designed to occupy the space between pure-chance games (baccarat) and skill-heavy games (poker): simple enough for a casual player to learn in one round, deep enough for regulars to develop a real strategy. Every round contains exactly seven decision phases, and every phase creates observable information - shown cards, spit targets, shine behavior - that skilled players can exploit.

Design lineage

Feature	Poker	Blackjack	Baccarat	Spit Shine
Decks	1	6-8	8	2
Players vs	Each other	Dealer	Banker	All seats
Hand construction	5 cards	2+ cards	2-3 cards	4 cards to 2
Decision points	4	1	0	7 phases
Player information	Partial	1 up card	None	Partial + Spit intel
Bluffing	Core	None	None	Show/hide + Bow Out
Card exchange	Draw poker	None	None	Spit phase
Discard / redraw	None	None	None	Shine phase
Safety valve	Fold	Surrender	None	Bow Out (25% refund)
House edge source	Rake	Rules	Commission	25% pot rake + dealer Shine

Unique innovations

- Spit: exchange one face-down card with an opponent. You see what you receive; they do not see what you gave. Each seat can be spat at once per round.
- Shine: optional redraw. Players discard 0-1 face-down cards; the dealer may discard 0-2 - the house's structural edge.
- Bow Out: hidden hands may fold and recover 25% of the ante, a risk-management lever that rewards discipline.
- Showing: revealing cards is a commitment (it locks out Bow Out) and an information weapon.

3. Game Modes & Player Configurations

Spit Shine supports two officially specified modes. Both use the same table geometry (up to four player seats plus the dealer) and the same seven-phase round, but the payout structures and the dealer's role differ. This section is the authoritative reference for how many players a table can seat, how empty seats are handled, and what each mode pays.

3.1 Competitive mode (default)

Competitive mode seats four players and the dealer at a full table; all seats ante within the 1-500 chip limits. The dealer is a full participant - the dealer can win the pot like any seat - but is barred from Bow Out and Spit, and may discard two cards in Shine instead of one. The highest-scoring hand at showdown wins 75% of the pot after the 25% rake.

Item	Specification
Seats	1-4 players + dealer (variable; full table = 5 seats)
Ante	Variable 1-500 chips per seat, including the dealer (examples below at 100)
Pot (full table)	500 chips
Winner receives	375 chips (pot - 25% rake of 125)
Dealer restrictions	No Bow Out, no Spit; may Shine 0-2 cards
Example	YOU wins with K+Q (350). Pot 500, rake 125, payout 375. YOU nets +275 chips for the round (100-chip ante).

3.2 Cooperative mode (players vs dealer)

Cooperative mode seats four players against the dealer. The dealer antes 0, so the pot is 400 chips. If any player beats the dealer at showdown, all non-bowed players share the 300-chip remainder after the 100-chip rake. If the dealer wins, the house keeps the entire pot. This mode's current economics are severely house-favorable and must be rebalanced before a casino deployment (see Section 14.5).

Item	Specification
Seats	1-4 players + dealer (variable)
Ante	Variable 1-500 chips per player (examples at 100); dealer antes 0
Pot (full table)	400 chips
Team win	All non-bowed players split 300 chips (75 each)
Dealer win	House keeps the full 400 chips
Audited team win rate	70.3% (20,000 rounds, seed 20260805)
Audited player EV	-46.1 chips per round (house edge ~46%)

Worked example: in a 10-round cooperative session, the team wins 7 rounds and the dealer wins 3. Each player's cash flow is $7 \times (-25)$ chips (pay 100, receive 75) plus $3 \times (-100) = -475$ chips, or -47.5 per round - consistent with the audited -46.1 before Bow Out refunds.

3.3 Player counts and seat configurations

Configuration	Seats	Use case	Notes
Heads-up (solo)	1 player + dealer	Single player vs dealer	Fully supported in both modes; competitive heads-up recommended (Section 3.5)
Live short-handed	2-3 players + dealer	Low-traffic hours	Drop in/out between rounds; empty seats do not ante
Live standard table	4 players + dealer	Recommended casino configuration	Players join/leave between rounds; capacity capped at 4
Digital POC / demo	1 human + 0-3 AI + dealer	Sales demo, training, simulation	AI fill optional; identical rules to live play

Tables are intentionally variable: players may drop in or out between rounds, and every seat may choose its own ante within the published 1-500 chip range. Seats never change mid-round; empty seats post no ante and take no cards. Every seat plays an independent hand, so the rules and odds per seat are identical at any table size - a short-handed table simply has a smaller pot. Casinos should publish the table minimums, cap the table at four player seats, and keep empty seats clearly marked.

3.4 Mode selection rules

- Mode is announced before the first ante of a session and cannot change mid-shoe.
- A table may run different modes on different shifts, but a single table cannot mix modes in one round.
- Side wagers are not part of the current rules; any future side bet requires independent math certification before deployment.

3.5 Solo vs dealer (heads-up)

A single player can play the dealer heads-up in both modes. The canonical engine and the Triple-A client support 1-4 player seats, and the heads-up table is the natural low-traffic configuration. The player's win rate is far higher than at a full table - they only need to beat the dealer - but the 25% rake still makes the game house-positive.

Metric (100-chip ante, 20k rounds)	Competitive heads-up	Cooperative heads-up
Player win rate	48.0%	43.5%
Dealer win rate	53.4%	56.5%
Player EV per round	-26.8 chips	-65.2 chips
House win per round	+49.4 chips (24.7% hold)	+65.2 chips (65.2% hold)
Pot	200 chips (both ante)	100 chips (dealer antes 0)
Verdict	Recommended solo configuration	House-favorable; not recommended as published

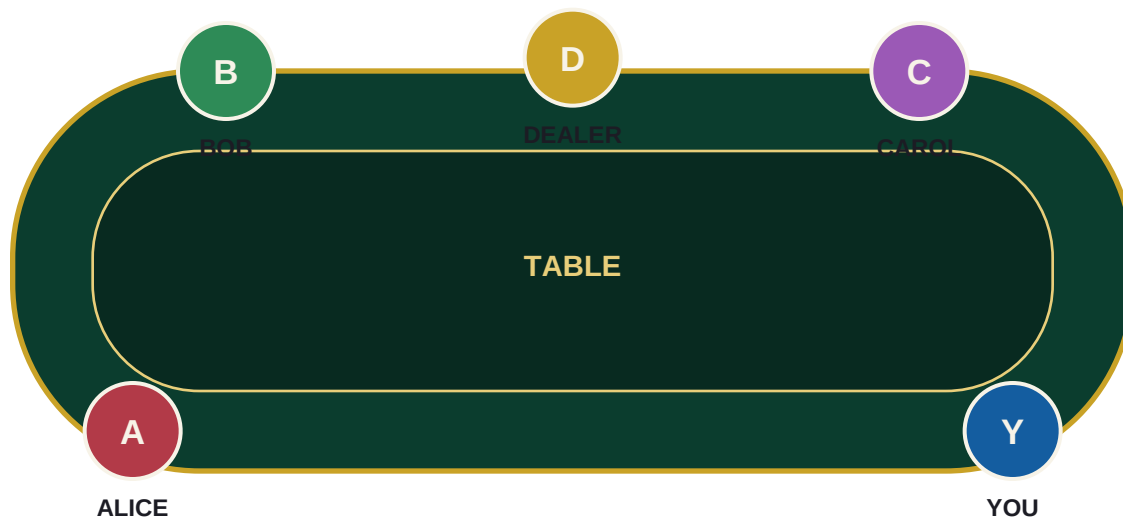
Recommended solo rule: competitive heads-up at a variable ante within table limits. The cooperative heads-up payout (player wins 75 chips against a 100-chip ante; dealer win forfeits the whole pot) gives the house a 65.2% hold and should be rebalanced or withdrawn, consistent with Section 14.5.

4. Equipment & Table Layout

Equipment checklist

Item	Specification	Control requirement
Cards	Two unopened 52-card decks, identical backs	Inspect for marks; count 52 per deck before and after play
Shoe	Dealing shoe for up to 104 cards	Tamper-evident; fixed position under surveillance
Cut card	Standard plastic divider	Insert 13-15 cards from the bottom (approx. 85-87% penetration)
Chips	Standard 25/100/500 denominations	House drop box for rake; chip fills recorded
Table	5 seats + dealer, 8-12 ft oval	Dealer station with shoe, drop box, discard rack
Surveillance	Overhead + dealer camera	Cards, shoe, and drop box visible at all times

Seating and deal order



Deal order (canonical): YOU, ALICE, BOB, CAROL, DEALER - one card at a time

The canonical engine deals round-robin one card at a time in seat order: YOU, ALICE, BOB, CAROL, DEALER. A floor operation may adopt standard clockwise dealing starting left of the dealer and dealing the dealer last; the two orders are statistically equivalent. The Spit turn order follows the same fixed rotation in the reference implementation and should be rotated per round on a live floor (Section 14.7).

5. Card Values & Scoring

Rank point values

Rank	Points	Rank	Points	Rank	Points
2	20	7	70	Q	150
3	30	8	80	K	200
4	40	9	90	A	100 or 0 (duality)
5	50	10	100	A+A	1000 (Double Aces)
6	60	J	100		

Ace duality

An Ace counts as 100 points or as 0 points, whichever maximizes the final hand score. $A+K = 300$; $A+2 = 120$; A alone paired with nothing is never forced to count 0 unless it is the better option. Two Aces together are the special Double Aces hand: 1000 points, the highest possible score. The maximum non-Ace pair is $K+K = 400$, which means no non-Ace hand can reach the 500+ range.

Scoring examples

Hand	Calculation	Score
K + Q	$200 + 150$	350
K + A	$200 + 100$ (Ace = 100)	300
10 + J	$100 + 100$	200
A + 2	$100 + 20$ (Ace = 100 beats 0 + 20)	120
7 + 5	$70 + 50$	120
A + A	Special rule	1000

Ace duality in practice - further worked examples

Hand	Pairings considered	Best score	Why
A K 7 2	$A+K=300$, $A+7=170$, $A+2=120$, $K+7=270$, $K+2=220$, $7+2=90$	300	Ace counts 100 with the King
A 2 3 4	$A+4=140$, $A+3=130$, $A+2=120$, $4+3=70$, $4+2=60$, $3+2=50$	140	Ace as 100 dominates every non-Ace pair
A 9 8 2	$A+9=190$, $A+8=180$, $A+2=120$, $9+8=170$, $9+2=110$, $8+2=100$	190	Ace + 9 beats 9 + 8 by 20 points
K K 5 3	$K+K=400$, $K+5=250$, $K+3=230$, $5+3=80$	400	The maximum non-Ace score
Q Q 10 J	$Q+Q=300$, $Q+J=250$, $Q+10=250$, $10+J=200$	300	Pair of Queens beats 10+J by 100

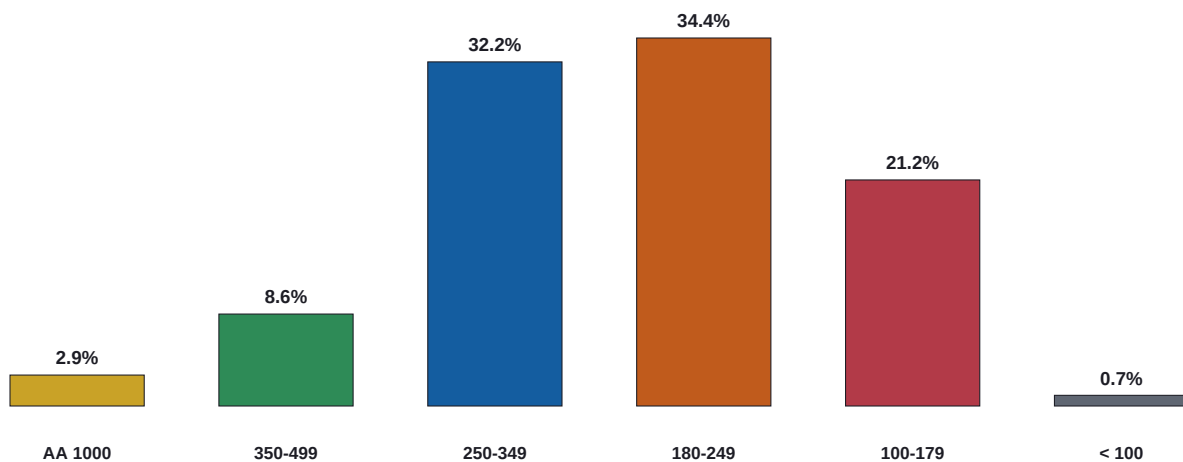
Hand	Pairings considered	Best score	Why
A A 2 3	A+A=1000 (special), A+2=120, A+3=130, 2+3=50	1000	Double Aces overrides all other pairings

Exact hand-score distribution (best 2 of 4 cards)

Tier	Exact count (of 4,598,126)	Probability	Note
Monster - Double Aces (1000)	133,126	2.90%	
500-999	0	0.00%	Impossible by rule - no non-Ace pair reaches 500
Strong (350-499)	395,702	8.61%	
Solid (250-349)	1,480,934	32.21%	
Marginal (180-249)	1,583,762	34.44%	
Weak (100-179)	972,856	21.16%	
Dead (< 100)	31,746	0.69%	

Source: exact enumeration of all $C(104,4) = 4,598,126$ four-card hands in the canonical engine. Suits affect only tie-breaks, so this distribution is exact for scores.

Exact hand-score distribution - best 2 of 4 cards



6. Hand Evaluation & Tie-Breaks

Evaluation algorithm

- Step 1: If the hand contains two Aces, the score is 1000 (Double Aces).
- Step 2: For every other hand, score each Ace as 100 or 0 to maximize the total.
- Step 3: The score is the sum of the two cards' points under that rule.
- Step 4: From four cards, evaluate all six possible pairings and keep the best.

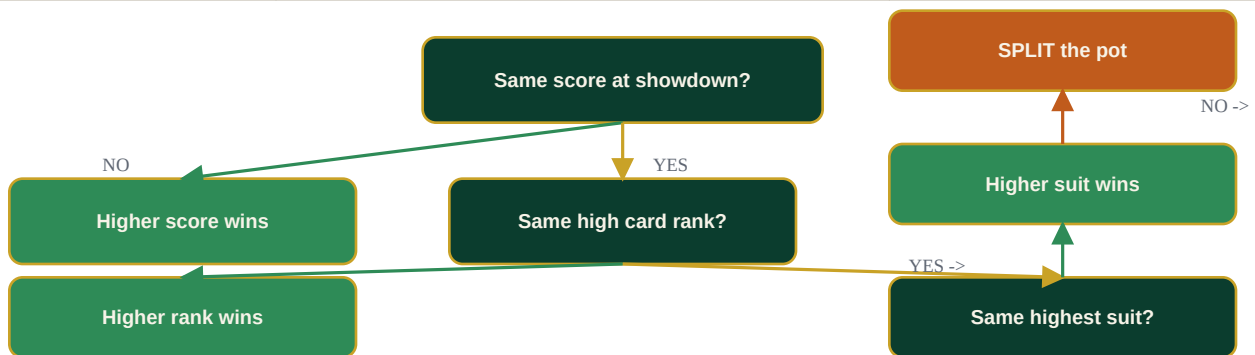
Showdown order (tie-breaks)

When two or more seats finish with the same score, the winner is resolved in this exact order: (1) higher score; (2) higher individual card rank; (3) higher suit value (clubs 1, diamonds 2, hearts 3, spades 4). If all three are identical, the pot is split equally among the tied seats.

Exact ties are not rare: 633 of 20,000 simulated rounds (3.16%) ended in a pot split. Any table procedure must define split payouts in advance (see Section 10.4).

Worked tie-break examples

Scenario	Hands	Resolution
Score tie, rank breaks it	ALICE: A+9 = 190 vs CAROL: 10+9 = 190	ALICE wins - high rank Ace (14) beats 10 (11)
Score and rank tie, suit breaks it	YOU: K(spades)+Q(clubs) = 350 vs ALICE: K(hearts)+Q(diamonds) = 350	YOU wins - highest suit value spades (4) beats hearts (3)
Full tie - pot splits	YOU: K(spades)+Q(hearts) = 350 vs ALICE: K(hearts)+Q(spades) = 350	Split - both score 350, rank 13, suit-max 4. Payout 375 becomes 187 + 187 + 1 chip rounding rake to the house



Audit note: earlier prototypes silently awarded the entire pot to the earliest seat on an exact tie. The canonical rule set adopted for casino operation splits the pot, matching the design documents.

7. How to Deal - Table Procedures

7.1 Pre-game preparation

- Open two new decks in front of surveillance; inspect every card for marks, bends, or irregularities; count 52 cards per deck.
- Shuffle: minimum seven riffle shuffles with alternating cuts, followed by a strip and a final riffle. Square the deck completely.
- Insert the cut card 13-15 cards from the bottom (recommended penetration 85-87% for a two-deck game).
- Load the shoe face-down, verify the deck's edge is even, and place the shoe in the marked position.
- Burn procedure: before the first round of a new shoe, draw one card face-down to the discard rack; it plays no part in the round.
- Announce table limits and confirm each seat's ante before dealing.

7.2 Round flow - dealer script

Step	Action	Dealer control
1	Collect antes: each active seat posts its declared ante within the published 1-500 chip limits (dealer posts an ante in competitive mode, 0 in cooperative).	Verify each ante against the table limits; place chips in the pot area; confirm the total.
2	Deal four cards to each seat, one card at a time, in seat order; dealer takes the last card of each pass.	No card may be exposed; announce 'Cards out' when complete.
3	Phase 1: allow private review. No table action.	Do not touch cards.
4	Phase 2: seats may show 0-4 cards. Announce each shown card as it is revealed; remind players that showing locks out Bow Out.	Record shown cards on the audit pad.
5	Phase 3: hidden seats may Bow Out. Pay 25% of ante (25 chips) and muck their cards.	Remove the refund from the pot; verify pot total.
6	Phase 4: each active non-dealer seat in turn may Spit once. Execute the swap without exposing the given card; reveal the received card to the spitter only.	Mark the target seat as 'spat' - one Spit per target per round.
7	Phase 5: seats may Shine (discard 0-1 face-down and draw). The dealer Shines last with 0-2 discards.	Draw replacements from the shoe; discard the old cards.
8	Phase 6: each seat selects its best two-card hand; the dealer selects their own pair.	Confirm no seat has more or fewer than four cards.
9	Phase 7: dealer reveals their own pair first, then each seat's pair; announce scores and determine the winner using score, rank, then suit.	On an exact tie, split the pot per Section 10.4.
10	Payout: rake 25% of the pot to the drop box; pay the winner(s); collect and count the discard rack.	When the cut card appears, finish the round and reshuffle.

7.3 Shoe and security controls

- The cut card ends the shoe: stop dealing after the round in which it appears, then reshuffle using the same procedure.
- The discard rack must be visible to surveillance at all times; discards are never returned to the shoe mid-shoe.
- A dealer change or table opening requires a full shuffle and a new burn card.

- Any exposed card during the deal is treated as a dead card: remove it to the discard rack and redeal that seat's card from the shoe.

7.4 Worked dealing examples

Example	Detail	Result
Shoe penetration	Cut card inserted 14 cards from the bottom of a 104-card shoe.	90 playable cards = 86.5% penetration. At ~22 cards per round (20 dealt + ~2 Shine draws), the shoe supports about four rounds before the cut card appears.
Shuffle sequence	Seven riffle shuffles with alternating cuts, a strip, and a final riffle, timed at roughly two minutes for a two-deck shoe.	Meets the minimum mixing requirement; the sequence is logged with the round audit for regulator review.
Dead card during deal	On the third pass, the dealer exposes the 5 of hearts while dealing to BOB.	The 5 of hearts is placed in the discard rack as a dead card and BOB receives the next card from the shoe. The round continues; no redeal.
Burn card	First round of a new shoe: one card is burned face-down to the discard rack before antes are collected.	The burned card plays no part in any round of that shoe.

8. How to Play - Phase-by-Phase Player Guide

Each round follows seven phases. The player guide below describes what happens, what actions are legal, and what the dealer does in each phase, followed by the strategy note used by the in-game coach.

Phase 1 - Private Reveal

Aspect	Detail
What happens	Four cards are dealt face-down to each seat. Only you can see your own cards.
Your options	No action required. Review your hand and form a plan.
Dealer's role	Dealer has no role beyond the deal.
Coach tip	Assess your best two-card pairing and classify it: Strong (350+), Solid (250-349), Marginal (180-249), Weak (below 180).

Phase 2 - Showing

Aspect	Detail
What happens	You may turn 0-4 of your cards face-up on the table. Shown cards are visible to every seat for the rest of the round.
Your options	Toggle any combination of face-down cards face-up, then confirm. Showing ANY card permanently locks out Bow Out.
Dealer's role	Dealer announces and records shown cards.
Coach tip	Strong hands often show one card to deter Spits; weak hands hide everything to keep Bow Out available. Showing is a commitment - never show a hand you are not willing to play.

Phase 3 - Bow Out

Aspect	Detail
What happens	If you showed no cards, you may forfeit the round and recover 25% of your ante (25 chips on a 100-chip ante).
Your options	Bow Out or stay. Bowing seats are removed from the round; their remaining pot contribution stays.
Dealer's role	Dealer processes the refund from the pot and mucks the cards.
Coach tip	Break-even math: staying costs 100, Bowing costs 75. Stay only if your win probability exceeds about 6.7%. Hands below roughly 150 points should normally Bow Out.

Phase 4 - Spit

Aspect	Detail
What happens	Each active non-dealer seat receives one Spit: give one face-down card to a target and take one of the target's face-down cards.

Aspect	Detail
Your options	Choose your card to give and your target. Each target can be spat at once per round. You may also pass.
Dealer's role	Dealer executes the swap, reveals the received card to the spitter only, and marks the target as spat.
Coach tip	Spit dumps your weakest card and draws a random replacement (expected value about 81 points). Simulations show Spitting is worth roughly 9 chips per round versus never Spitting. Strong hands (350+) usually pass to avoid disruption.

Phase 5 - Shine

Aspect	Detail
What happens	You may discard 0-1 face-down cards and draw replacements from the shoe. The dealer may discard 0-2.
Your options	Choose a face-down card to discard, or keep your hand. You draw only if you discard.
Dealer's role	Dealer discards and draws for each seat, then for the dealer seat.
Coach tip	Below 180 points, Shine is usually correct. Above 250, keep. The dealer's extra discard is the house's structural edge - plan around it.

Phase 6 - Selection

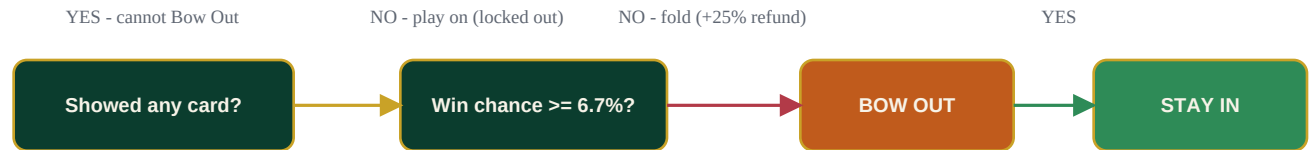
Aspect	Detail
What happens	From your four cards, select the two-card pairing you will play at showdown.
Your options	Choose any two cards. The selection is final.
Dealer's role	Dealer selects the dealer pair.
Coach tip	Always consider all six pairings. The engine recommends the maximum score; remember Aces count 100 or 0 to maximize.

Phase 7 - Showdown & Payout

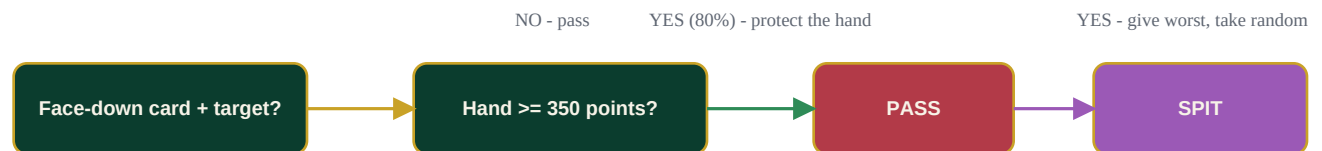
Aspect	Detail
What happens	All remaining seats reveal their pairs. The highest score wins; ties split.
Your options	No player action - results are announced.
Dealer's role	Dealer reveals first, then each seat; announces scores, applies tie-breaks, rakes 25%, and pays the winner(s).
Coach tip	Watch the dealer's revealed pair every round - it is the single most informative datum at the table.

Decision guides (reference cards)

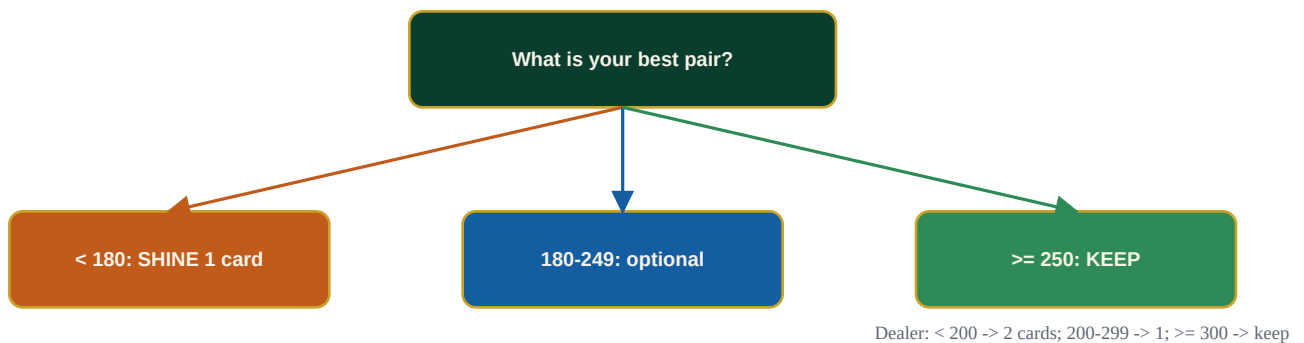
Bow Out decision



Spit decision



Shine decision



9. Roles & Responsibilities

A Spit Shine table runs on three human roles: the Player, the Dealer, and the Pit Manager (shift supervisor). The rules are identical for every seat; the roles define who does what, who is responsible for what, and what each role must watch for to keep the game fair. The Pit Manager's watchlist is the game's primary defense against cheating, collusion, and procedural error.

Responsibility matrix

Duty	Player	Dealer	Pit Manager
Post ante within limits	Primary	Verify	Set limits
Shuffle / shoe / cut card	-	Primary	Observe
Deal and phase execution	-	Primary	Spot-check
Announce shown cards and scores	-	Primary	Audit log
Rake and payouts	-	Primary	Verify + drop
Dispute resolution	Rights	Escalate	Final
Fairness monitoring	Play by rules	Self-check	Primary

9.1 The Player

The player's job is to make seven decisions per round under information asymmetry: what to show, whether to stay or fold, whether to exchange a card, whether to redraw, and which two cards to play. The full decision guide is in Section 8.

Player duty	Detail
Session setup	Buy in at least 20x the table ante; choose an ante within limits; confirm the table minimums and the mode before play
Hand protection	Keep all cards above the table and in view; never touch another seat's cards or the pot
Phase discipline	Decide Showing, Bow Out, Spit, Shine, and Selection only when invited; a confirmed action is final
Information	The Spit-received card is private to you; sharing it with other players is against house rules
Complaints	Never argue at the table; call the Pit Manager or request the audit log

Player example: a \$1,000 buy-in at a 100-chip ante covers the recommended 20-round cushion. Two wins, six losses, and two Bow Outs over ten rounds nets -236 chips (worked in Section 12.6).

9.2 The Dealer

The dealer executes the game under Pit Manager observation. Full dealing procedures are in Section 7 and payout math in Section 10; this subsection states the dealer's standing duties and self-checks.

Dealer duty	Detail
Equipment control	Verify 52 cards per deck, inspect for marks, shuffle seven riffles + strip + final riffle, insert the cut card, burn one card per new shoe
Ante control	Verify every ante against the published limits before dealing; announce the pot total

Dealer duty	Detail
Deal control	Round-robin one card at a time; never expose a card; handle dead cards per Section 16.2
Phase control	Announce each phase; confirm shows; process Bow Out refunds (25% of ante, floored); execute Spit swaps without exposing the given card; control Shine discards and draws
Payout control	Compute rake (25% of pot), apply tie-breaks, announce scores, pay winners, and move the rake to the drop box before any other action
Audit discipline	Never settle a payout or dispute without the audit log; count the tray at shift start and end

Dealer self-checks every round: card count per seat, face-down integrity during Spit, one target per Spit, discard limits (players 0-1, dealer 0-2), and pot = antes minus refunds. Any discrepancy stops the round and summons the Pit Manager.

9.3 The Pit Manager

The Pit Manager owns fairness, money flow, and dispute resolution. The manager inspects the table at open and close, observes shuffles and payouts, reviews the audit log, controls the drop box, and maintains the published limits.

What can make the game unfair - the watchlist

Category	Watch for	Why it matters
Equipment	Marked, bent, or trimmed cards; missing cards; duplicate mismatches; patterned or insufficient shuffles; misplaced cut card	Compromised randomness or known-card play
Player collusion	Two seats repeatedly Spitting at each other; players sharing private Spit information; signaling; chip passing or soft play between partners	Transfers EV and leaks private information
Betting abuse	Past posting (adding chips after the result); late antes; ante jumps correlated with deck composition; oversized or phantom bets	Direct theft and advantage betting
Card handling	Face-down cards touched or re-arranged; cards mucked early; hands altered after reveal	Card switching and hand manipulation
Dealer errors	Payout or rake mistakes; exposed cards; wrong discard counts; Spit swaps that reveal the given card; unannounced shown cards	Unfair or uncertified play, disputes
Procedural	Mid-round seat changes; antes outside limits; missed burn or cut-card reshuffle; unlogged payouts	Breaks the certified procedure and audit trail

Phase-by-phase pit watch

Moment	What the Pit Manager watches
Before deal	Shuffle and cut; ante limits; roster matches the seated players; shoe position
Deal	Card counts; exposures; deal order; burn card
Showing	Announced cards match the physical cards; no re-hiding after confirmation
Bow Out	Refunds equal 25% of each ante; pot updated; mucked cards untouched
Spit	Swap executes without exposing the given card; the received card is shown only to the spitter; one target per round; repeated same-pair Spitting across rounds
Shine	Discard limits (0-1, dealer 0-2); a draw only after a discard; no shoe fishing
Selection	No changes after confirmation

Moment	What the Pit Manager watches
Showdown & payout	Scores and tie-breaks announced; rake to drop; payout correct; audit entry written

Pattern cues for surveillance: the same two seats exchanging Spits round after round, a player who always Bows Out when a partner stays, chip movements that do not match payouts, and antes that jump exactly when the shoe runs long.

9.4 Suspected-problem procedure

- Observe and document; never accuse a player at the table.
- Let the round finish, then notify surveillance and the floor.
- Review the audit log and the cameras; decide by evidence.
- Apply house policy (warning, exclusion, or regulator report) without disturbing the game.

10. Dealer Procedures & Payout Math

9.1 Pot accounting

The pot is the sum of all antes minus Bow Out refunds. At showdown the house rakes 25% (floor of the integer division) and the remainder is paid to the winner, or split among tied winners. Any chips left over after an equal split (for example, splitting 375 two ways leaves 1 chip) are retained by the house as rounding rake - this rule must be posted at the table.

9.2 Payout examples

Scenario	Pot	House rake (25%)	Winner(s)	Payout
Full table, no Bow Outs	500	125	One winner	375
One Bow Out (25 refund)	475	118	One winner	357
Two Bow Outs (50 refunds)	450	112	One winner	338
Exact tie, full table	500	125	Two winners	187 each + 1 rounding rake
Cooperative, players win	400	100	Four players	75 each (300 total)
Cooperative, dealer wins	400	-	Dealer	House keeps entire pot (400)

Rounding rake note: $500 \times 25\% = 125$, leaving 375. Splitting 375 between two winners yields $187 + 187 + 1$ remainder, which is retained by the house and logged. All payouts must be announced before chips move.

9.3 Dealer's own play

The dealer is a full seat in competitive mode (posts an ante like any seat - 100 chips in the audit model - and can win the pot) but never Bows Out and never Spits. The dealer's structural advantage is the two-card Shine: in 20,000 simulated rounds the extra discard is worth roughly 6.8 points of win rate, lifting the dealer to a 24.6% win rate versus a fair 20%. Dealers follow the same thresholds as the reference AI: Shine 2 cards below 200 points, 1 card at 200-299, keep at 300+.

9.4 Disputes and corrections

- Any exposed card during the deal is a dead card: discard it and draw a replacement for that seat.
- If a player exposes a face-down card during Spit or Shine, the card is still played (it becomes known information); the floor decides penalties per house policy.
- Payout disputes are resolved from the audit log; every round's deal, show, bow, spit, shine, selection, and payout events are recorded with card-level detail.

9.5 Worked payouts with chip values

Chip value	Ante	Pot after one Bow Out	Rake (25%)	Winner payout
\$0.25	\$25	475 chips = \$118.75	118 chips = \$29.50	357 chips = \$89.25
\$1.00	\$100	475 chips = \$475.00	118 chips = \$118.00	357 chips = \$357.00
\$5.00	\$500	475 chips = \$2,375	118 chips = \$590	357 chips = \$1,785

Worked split example at \$1 chips: full table, two winners split 375. Each winner receives 187 chips (\$187); the remaining 1 chip (\$1) goes to the house as rounding rake and is recorded in the audit log.

11. Limits, Bankroll & Money Management

Tables do not have fixed stakes: the ante is variable per seat within the 1-500 chip range, and players may drop in or out between rounds. This section defines the chip denominations, the table limits, the buy-in and bankroll guidance, and the cash and chip controls a floor operation needs.

11.1 Chip denominations and ante limits

Denomination	Value at \$1/chip	Typical use	Buy-in guidance (20x ante)
1 chip	\$1	Minimum ante unit	20 chips = \$20 at ante 1
5 chips	\$5	Low-stakes play	100 chips = \$100 at ante 5
25 chips	\$25	Standard low limit	500 chips = \$500 at ante 25
100 chips	\$100	Standard table ante	2,000 chips = \$2,000 at ante 100
500 chips	\$500	High limit	10,000 chips = \$10,000 at ante 500

Ante limits: minimum 1 chip, maximum 500 chips per seat. The house publishes its table minimum and maximum inside this range; each seat chooses its ante freely within those limits each round. The maximum single-round win is the full pot minus rake - at a full table at 100-chip antes that is 375 chips. A player can never win more than the pot, so no separate jackpot exposure exists unless a future side wager is added (which would require its own certification).

Variable table rules

- Players may join or leave between rounds; seats never change mid-round, and empty seats post no ante and take no cards.
- Table capacity is four player seats plus the dealer; the table publishes its minimum player count (one for heads-up).
- Ante spread limit (recommended): the largest and smallest ante at one table should stay within a 5x spread. Mixed antes transfer EV toward smaller bettors (Section 14.9), so an unbounded spread is bad for high-ante players and for the game's perceived fairness.

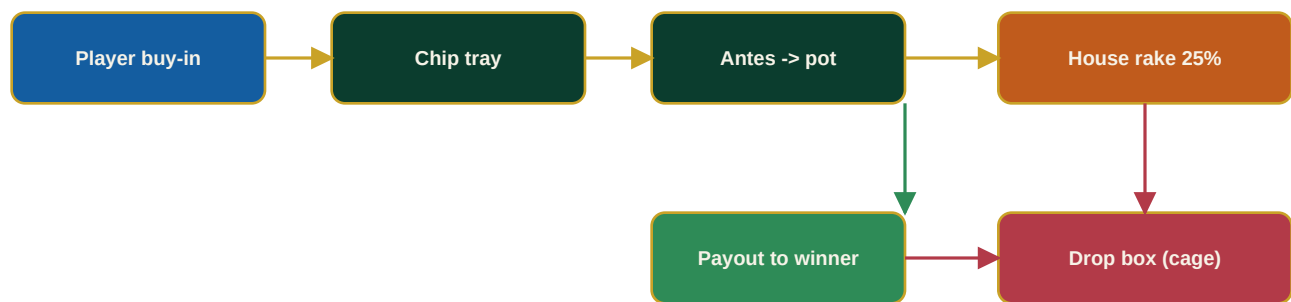
11.2 Player bankroll guidance

Guideline	Rule of thumb	Worked example (\$1 chips)
Session buy-in	At least 20x the ante	100-chip ante -> \$2,000 buy-in; 25-chip ante -> \$500
Session loss limit	Stop at 5x the ante	Stop-loss at 5x your ante per session
Bow Out discipline	Fold below ~150 points when hidden	Saves 25 chips per fold; at ~9% of rounds this adds ~2.3 chips per round versus never folding
Expected round cost	Audited EV scales with ante: about -0.34 chips per chip wagered at a full table (seat 0)	At a 100-chip ante, EV -33.9/round; a \$2,000 bankroll supports roughly 50-60 rounds on average

11.3 Cash and chip flow

Funds move in a closed loop: player buy-ins exchange cash for chips from the tray; antes move chips to the pot; at showdown the rake moves to the drop box and the payout moves back to the winner's stack; Bow Out refunds return to the player. At end of shift the chip equation must balance.

Reconciliation Item	Amount	Worked example
Opening float (tray)	+\$2,000	Chips issued to the table
Player buy-ins (cash)	+\$4,000	Cash exchanged for chips
Closing tray	-\$2,100	Chips still in the tray
Drop box (cash removed)	-\$3,900	Cash moved to the cage
Check	$\$2,000 + \$4,000 = \$2,100 + \$3,900$	Balanced: $\$6,000 = \$6,000$
Audit theoretical win	\$1,100	Sum of per-round rakes recorded by the audit log
Variance monitoring	+2.9%	Actual win \$1,132 vs theoretical \$1,100 - within the standard +/-3% tolerance; investigated only if outside



Bow Out refunds return 25% of ante to the player before rake is computed

11.4 Responsible gaming and jurisdictional limits

- Publish table minimums and maximums at the table and in the house rules; they are configurable per jurisdiction and per venue.
- Support player-set loss limits, time limits, and voluntary exclusion through the standard responsible-gaming tools.
- Currency transaction reporting and AML thresholds apply at the cage, not the table; the dealer never handles cash directly - all buy-ins are exchanged through the tray under surveillance.

12. Daily Turnover & Revenue Model

This section models the funds that flow through a Spit Shine table and the revenue it can generate. All figures are illustrative and depend on four operator-set variables: chip value, round duration, table occupancy, and operating hours. The audit supplies the fixed part: the house keeps 24.5% of the pot (rake) in competitive mode.

12.1 Definitions

Term	Definition	Spit Shine value
Handle (gross turnover)	Total antes placed across all seats in a period	Audited average 565 chips per round under the variable-table model (Section 14.9)
Table win (hold)	House revenue before expenses	Rake only: 24.5% of pot in competitive mode
Hold percentage	Table win / handle	24.5% audited, competitive, at any table size or ante
Drop	Cash and chips exchanged at the table	Tracked per shift; reconciled against theoretical win
Rounds per hour	Dealt rounds per hour of operation	Model range 17-30 (see assumptions)

12.2 Model assumptions (must be validated in floor trials)

Variable	Conservative	Standard	Aggressive
Round duration	3.5 minutes	2.5 minutes	2.0 minutes
Rounds per hour	17	24	30
Operating hours / day	12	12	24
Table model	Variable-table audit model	Variable-table audit model	Variable-table audit model
Chip value	\$1	\$1	\$1
House win per round	\$138.40 (audited average)	\$138.40	\$138.40

12.3 Per-round economics

Configuration	Handle per round	Expected house win	Winner payout
Variable table (audited model, avg 3.59 players, mixed antes)	565 chips = \$565	\$138.40 (24.5%)	Varies with pot
Competitive, full table at 100-chip antes (4 players + dealer)	500 chips = \$500	\$122.50 (24.5%)	375 chips = \$375
Solo heads-up at 100-chip antes (1 player + dealer)	200 chips = \$200	\$49.40 (24.7%)	150 chips = \$150
Competitive, 3 players + dealer	400 chips = \$400	\$98.00 (24.5%)	300 chips = \$300
Cooperative, full table (4 players, dealer antes 0)	400 chips = \$400	\$184.40 (46.1% - flagged)	75 chips per winning player

12.4 Daily turnover scenarios (worked)

Scenario	Rounds per day	Handle	House win (24.5%)
Conservative: 17 rph x 12 h	204	204 x \$565 = \$115,260	\$115,260 x 24.5% = \$28,215
Standard: 24 rph x 12 h	288	288 x \$565 = \$162,720	\$162,720 x 24.5% = \$39,833
Aggressive: 30 rph x 24 h	720	720 x \$565 = \$406,800	\$406,800 x 24.5% = \$99,582

These figures use the audited variable-table model directly: the average table fields 3.59 player seats with mixed 1-500 chip antes, so no separate occupancy factor is needed. Handle scales linearly with average ante: at a 25-chip average ante, divide by 4; at a 500-chip average ante, multiply by 5.

12.5 Monthly and multi-table projections

Deployment	Handle / day	House win / day	House win / 30 days
1 table, standard	\$162,720	\$39,833	\$1,194,990
5 tables, standard	\$813,600	\$199,165	\$5,974,950
10 tables, standard	\$1,627,200	\$398,330	\$11,949,900

These projections assume the model's standard case (24 rounds/hour, 12 hours/day) and full-year operation at the modelled drop-in/out roster. They are deliberately striking because a 25% rake is far above mainstream table games (blackjack hold is typically 10-25% of drop but with much lower edge per bet; baccarat holds 10-15%). The rake is the game's key commercial lever: Section 12.7 shows calibration options.

12.6 Player session example (funds flow)

Round	Result	Player cash flow	Running total
1	Win (full table)	+257	+257
2	Loss	-100	+157
3	Bow Out	-75	+82
4	Loss	-100	-18
5	Win (full table)	+257	+239
6	Loss	-100	+139
7	Bow Out	-75	+64
8	Loss	-100	-36
9	Loss	-100	-136
10	Loss	-100	-236

Session summary: \$1,000 buy-in, 10 rounds, 2 wins, 6 losses, 2 Bow Outs. Net = 2 x (+257) + 6 x (-100) + 2 x (-75) = +514 - 600 - 150 = -236 chips (-\$236 at \$1 chips). This is an illustrative sequence, not a prediction.

12.7 Calibration options for market fit

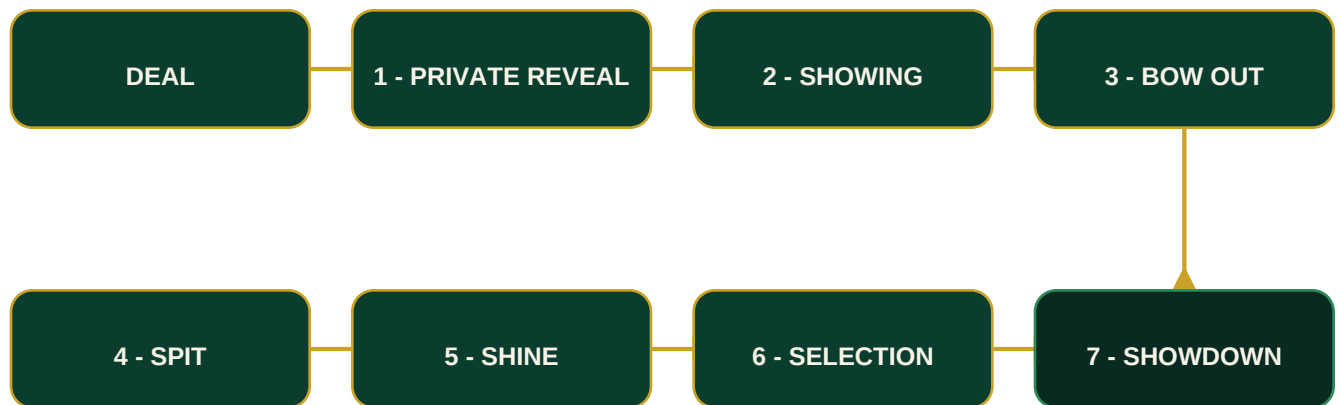
Rake	Hold (approx.)	House win per full round	Note
25% (current)	24.5%	\$122.50	Audited baseline; very high for a table game

Rake	Hold (approx.)	House win per full round	Note
15%	14.7%	\$73.50	Closer to premium table-game norms
10%	9.8%	\$49.00	Competitive with popular carnival games

The operator chooses the rake and chip value as a package. A lower rake improves player retention and table longevity; a higher chip value restores revenue per table. Both variables must be settled before the independent game-math certification, because they change the certified payout tables.

13. Game State Machine & Canonical Rules

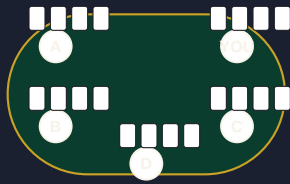
Round state flow



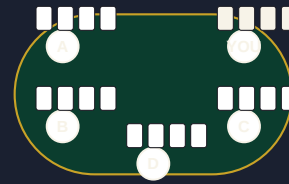
The table in every state

1 - DEAL

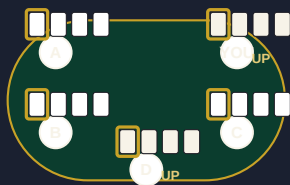
4 cards each, face-down

**2 - PRIVATE REVEAL**

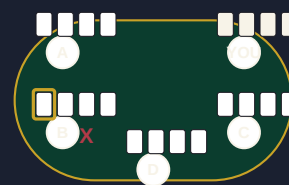
YOU see your hand

**3 - SHOWING**

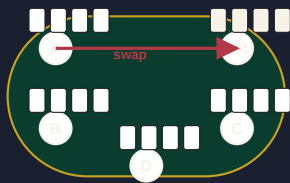
cards may be shown; locks Bow Out

**4 - BOW OUT**

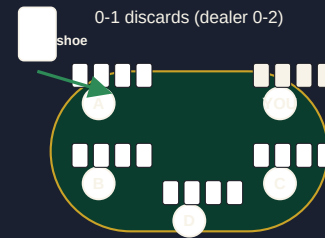
hidden seats may fold (+25%)

**5 - SPIT**

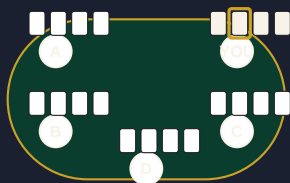
give 1 / take 1; each target once

**6 - SHINE**

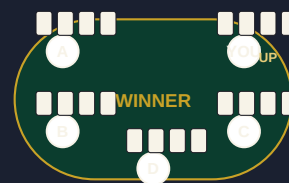
0-1 discards (dealer 0-2)

**7 - SELECTION**

best 2 of 4 cards

**8 - SHOWDOWN**

score -> rank -> suit -> split



Each panel shows the table state at one moment of the round: face-down cards are dark, face-up cards are light, gold rings mark shown or selected cards, X marks a Bow Out, and arrows show card movement. The same states are reflected in the audit log after every round.

Canonical rules reference (v11)

Rule	Specification
Shoe	104 cards (2 decks), fresh and shuffled every round
Deal	4 cards per seat, round-robin, one card at a time
Ante	100 per seat; dealer antes 0 in cooperative mode
Showing	0-4 cards; showing any card locks out Bow Out
Bow Out	Hidden seats only; refund 25% of ante; dealer never bows
Spit	One per non-dealer seat; each target once per round; received card private to spitter
Shine	0-1 discards (players), 0-2 (dealer); draw only after discard
Selection	Best 2 of 4; human may choose any two cards
Showdown	Score, then high rank, then high suit; exact ties split
Payout	Rake 25% of pot; winners share remainder; rounding rake to house
Modes	Competitive (all seats) and Cooperative (players vs dealer)

State invariants

- A seat can Bow Out only once and only while hidden.
- A seat can Spit at most once; a target can receive at most one Spit.
- Spit and Shine operate only on face-down cards.
- The pot never goes negative; refunds are taken from the pot before rake.
- The dealer is always active at showdown (never Bows Out).
- Every action is written to the audit log before the next action begins.

14. Audit Data - Odds, EV & Fairness

All figures below are produced by the canonical engine v11 and the audit harness (spitshine_analysis.py). Exact probabilities are enumerated combinatorially; round-level figures come from 20,000 Monte Carlo rounds per scenario with a fixed seed (20260804), playing the documented threshold policies at every seat. Chip conservation (sum of seat nets + house take = 0) holds to machine precision.

14.1 Exact probabilities

Event	Probability	Method
A specific 2-card pair is A+A	0.52%	Exact - 28 of $C(104,2) = 5,356$ pairs
Your 4-card hand contains A+A	2.90%	Exact hypergeometric
At least one seat dealt A+A in a round	13.92%	Monte Carlo - 200,000 deals
At least one seat finishes with A+A after Spit/Shine	17.2%	20,000-round simulation
Best pair 350+ (any seat's deal)	11.50%	Exact hand enumeration

The Spit and Shine phases create roughly 3.3 points of additional Double-Aces frequency because a lone Ace can be paired by a redraw or exchange. This is a deliberate feature of the game's variance profile.

14.2 Round outcomes - competitive mode (20,000 rounds)

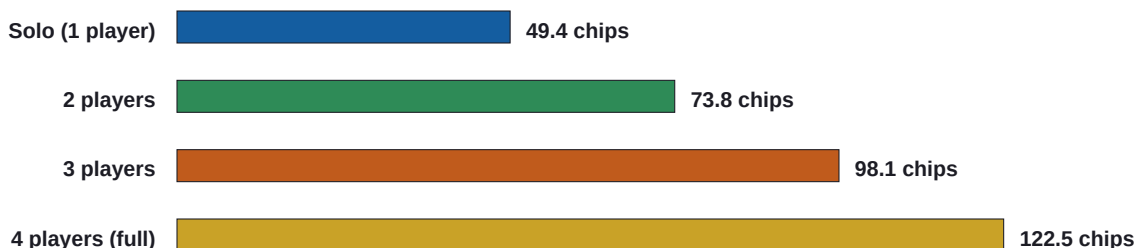
Seat	Win rate	Avg final score	Net EV per round
YOU (seat 0)	17.8%	241.5	-33.9
ALICE (seat 1)	19.3%	246.7	-28.7
BOB (seat 2)	20.0%	249.0	-26.5
CAROL (seat 3)	21.4%	255.3	-21.0
DEALER (seat 4)	24.6%	278.0	-12.5
House (rake)	-	-	+122.5 per round

Win rate by seat - competitive (fair baseline 20%)



Every seat - including the dealer - has negative expected value in competitive mode; only the house rake is positive. The dealer's 24.6% win rate is the strongest structural advantage in the game, and it comes almost entirely from the two-card Shine.

House win per round by table size (100-chip ante, audited)



House win scales with the pot: 24.5% of (players + dealer) x ante. The hold percentage is invariant; only the volume changes with table size.

14.3 Winner score distribution

Winning score	Rounds	Share
Double Aces (1000)	3,442	17.2%
350-499	7,307	36.5%
250-349	8,777	43.9%
180-249	474	2.4%

14.4 House edge - corrected mathematics

With a full 500-chip pot and a fair 20% win rate, the winner nets $375 - 100 \text{ ante} = +275$ chips. Expected value is therefore $E = 0.20 \times 275 - 0.80 \times 100 = -25$ chips per round. That is a 25% house edge per ante, exactly the 25% rake

- not the 5% claimed in the original white paper, which incorrectly treated the gross 375 payout as profit. Measured house take in simulation is 122.5 chips per round (24.5%) after Bow Out refunds.

Break-even question	Correct answer	Formula
Stay in vs Bow Out	~6.7% win chance	25 / 375
Break even for the round	~26.7% win chance	100 / 375
Fair-game EV (20% win rate)	-25 chips per round	$0.2 \times 275 - 0.8 \times 100$

Worked EV calculations

Perspective	Formula	Result	Measured (20k rounds)
Player, seat 0	$0.1783 \times 275 - 0.8217 \times 100$	-33.1 chips / round	-33.9 chips / round
Dealer seat	$0.2461 \times 275 - 0.7539 \times 100$	-7.7 chips / round	-12.5 chips / round
House	Rake only (25% of pot)	+125.0 chips / round (full pot)	+122.5 chips / round
Bow decision, 12% win chance	$0.12 \times 275 - 0.88 \times 100$	-55.0 -> stay (bow costs -75)	-
Bow decision, 5% win chance	$0.05 \times 275 - 0.95 \times 100$	-81.3 -> bow (saves 6.3)	-

The small gap between formula and measured EV comes from pot-size variation: Bow Out refunds shrink the pot, so the actual winner payout is below the 375-chip full-pot figure used in the closed-form formula.

14.5 Cooperative mode economics and rebalancing

In cooperative mode the dealer antes 0, the pot is 400, and the team of four players wins if any player beats the dealer. The team wins 70.3% of rounds in simulation, yet each player still nets -46.1 per round because a team win pays only 75 chips gross per player, and a dealer win forfeits the entire pot. The house takes 184.4 chips per 400-pot (46.1% effective edge). This mode is not viable for a casino floor in its current form and is flagged for rebalancing.

Rebalance package (illustrative, pending certification)

Option	Change	Audited/derived effect
A - Reduce dealer Shine in cooperative	Dealer discards 0-1 cards like players	Team win rate rises from 70.3% to 78.6% (20,000-round experiment, seed 20260806)
B - Dealer antes + higher team payout	Dealer antes 100; team win pays 125 gross per player (pot 500)	Player EV improves from -46.1 to about -12.1 chips/round
C - Combined package	Both A and B	Player EV about -1.8 chips/round - near fair, comparable to standard house-banked games

Option C is the recommended starting point for a casino deployment: it removes the dealer's double-discard edge in this mode and makes the payout structure fair while keeping the house positive. It must be formally simulated and certified before publication.

14.6 Rule variants - what each mechanic contributes

Variant	Dealer win	YOU win	YOU net/round
Baseline (canonical)	24.6%	17.8%	-33.9

Variant	Dealer win	YOU win	YOU net/round
Dealer shines 1 card (like players)	18.3%	18.9%	-30.1
Dealer also Spits (legacy v4 rule)	29.7%	17.1%	-36.1
Spit order rotates each round	24.9%	18.9%	-30.3
Human never Spits	26.1%	15.3%	-42.9
Nobody Shines	17.8%	18.8%	-30.6
Conservative 'smart' human	25.1%	15.3%	-36.3

Reading the variants: the dealer's second Shine discard is worth about 6.8 points of win rate (24.6% to 18.3%); the legacy v4 rule that let the dealer Spit pushed the dealer to 29.7%; rotating the Spit order equalizes the player seats; and never Spitting costs the player roughly 9 chips per round.

14.7 Seat-position fairness

With a fixed Spit order, seat 0 (YOU) wins 17.8% while seat 3 (CAROL) wins 21.4% - a systematic positional bias of about 3.7 points. Reversing the order inverts the bias, confirming the Spit sequence itself is the cause. The recommended floor fix is to rotate the Spit starting seat each round (poker-button style), which equalizes the four player seats without changing house economics. This must be included in the table procedures.

14.8 Solo vs dealer (heads-up) odds

The canonical engine seats 1-4 players plus the dealer, making a true heads-up table possible. The player only needs to beat the dealer, so heads-up win rates are far higher than at a full table; the 25% rake still leaves the house positive in competitive mode.

Metric (100-chip ante, 20k rounds)	Competitive heads-up	Cooperative heads-up	Full table (reference)
Player win rate	48.0%	43.5%	17.8%
Dealer win rate	53.4%	56.5%	24.6%
Player EV / round	-26.8	-65.2	-33.9
House win / round	+49.4 (24.7% hold)	+65.2 (65.2% hold)	+122.5 per round
Player average final score	261.8	266.1	241.5

Competitive heads-up is the recommended solo configuration. Cooperative heads-up gives the house a 65.2% hold and should be rebalanced or withdrawn (see Section 14.5).

14.9 Variable-table simulation (drop-in/out, mixed antes)

To model a real table, 20,000 rounds were simulated with players dropping in and out between rounds (10% leave probability, 45% join probability, capacity 4) and every seat's ante sampled per round from a mixed 1-500 chip distribution (55% low, 30% mid, 15% high). The dealer posts a fixed 100-chip ante in competitive mode.

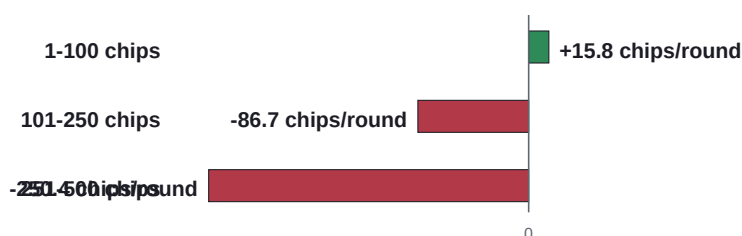
Metric	Value	Note
Average player seats	3.59	1p: 1.3% of rounds, 2p: 7.3%, 3p: 22.6%, 4p: 68.8%
Drop-ins / drop-outs	5,156 / 5,154	Balanced over 20,000 rounds

Metric	Value	Note
Handle per round	565.3 chips	Mixed antes across 3.59 average seats + dealer
House win per round	138.37 chips	Hold 24.48% - invariant to table size
Player (YOU) net	-54.5 per round	Reflects the mixed ante model, not a single ante level
Dealer seat net	+6.1 per round	Fixed 100 ante; benefits from winning mixed-stake pots

EV transfer across ante bands (audited)

YOU ante band	Rounds	Net EV per round
1-100 chips	11,005	+15.8
101-250 chips	6,038	-86.7
251-500 chips	2,957	-250.4

YOU EV by ante band (variable-table audit, 20k rounds)



The winner takes 75% of the whole pot regardless of their own ante, so small bettors at a mixed-stakes table capture outsized pots and can be net positive, while large bettors lose more than their proportional share of the rake. This is a real design consequence of variable stakes: operators should publish an ante spread limit (recommended 5x between the smallest and largest ante at one table) or run uniform-ante tables to protect high-ante players.

14.10 One-million-round verification

To confirm every headline figure at casino-grade sample size, the canonical engine ran 1,000,000 rounds for each of the three flagship scenarios (competitive full table, variable drop-in/out table, and solo heads-up) with audit logging disabled for speed and fixed seeds. Total compute was about 15 minutes. Win rates carry a 95% confidence interval of roughly +/-0.08 to +/-0.10 percentage points.

Scenario	Metric	1M result	95% CI	20k reference
Competitive	YOU win rate	17.73%	+/-0.07 pp	17.83%
Competitive	Dealer win rate	24.66%	+/-0.08 pp	24.61%
Competitive	YOU EV per round	-34.42 chips	-	-33.88
Competitive	House win per round	122.53 chips	-	122.51
Variable table	Average players	3.586	-	3.589
Variable table	Hold percentage	24.47%	-	24.48%

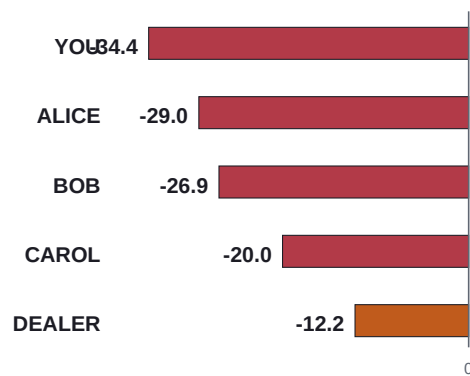
Scenario	Metric	1M result	95% CI	20k reference
Variable table	House win per round	138.27 chips	-	138.37
Solo heads-up	YOU win rate	48.48%	+/-0.10 pp	47.98%
Solo heads-up	YOU EV per round	-25.84 chips	-	-26.82

Every 20,000-round figure survives the million-round check inside the expected sampling error. The verified constants: hold 24.47% (variable table), competitive house win 122.53 chips per round at 100-chip antes, solo player win rate 48.49%, and the seat-order bias of about 4 points between seat 0 and seat 3.

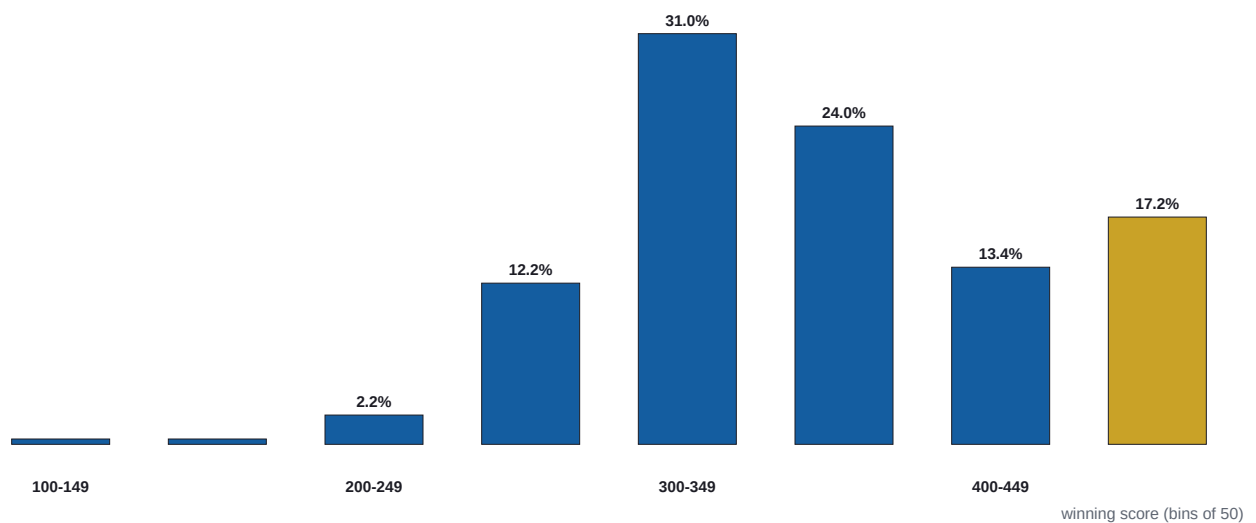
Win rate by seat - competitive, 1,000,000 rounds



Net EV per seat per round - competitive, 1,000,000 rounds (100-chip ante)

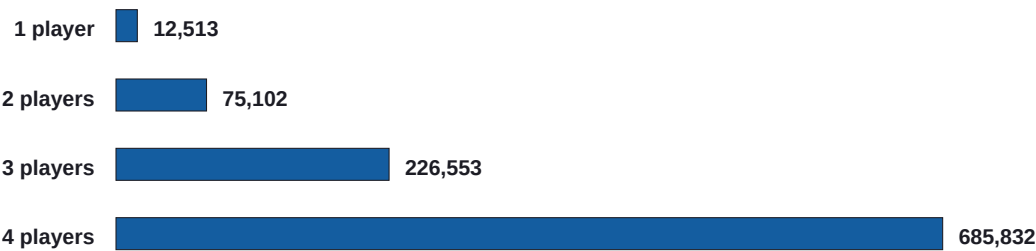


Winning-score distribution - competitive, 1,000,000 rounds

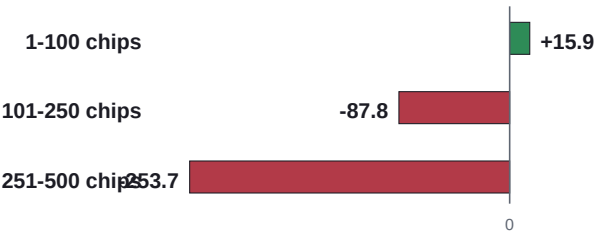


Variable-table metrics - 1,000,000 rounds

Rounds by table size - variable-table model



YOU EV by ante band - variable-table model, 1,000,000 rounds



Solo heads-up win rate - 1,000,000 rounds



The EV transfer across ante bands is confirmed at 1,000,000 rounds: 1-100 chip bettors net +15.9 chips per round while 251-500 chip bettors lose -253.7 - the strongest argument for the 5x ante spread limit.

14.11 Two-million-round hand audit & 24-hour distribution

The canonical engine played 2,000,000 competitive rounds while tracking every seat's hand and point totals per round: the dealt best pair, the final selected pair, and the final score for all 10,000,000 hands. The full-run aggregates are below; a raw 5,000-round sample (every card, every score) is exported to spitshine_audit_2m_sample.csv.

Audit metric	Value
Rounds played	2,000,000
Hands tracked	10,000,000 (8,000,000 player hands)
Total handle	1,000,000,000 chips (500 per round at 100-ante)
House win total	245,057,169 chips (122.53 per round)

Audit metric	Value
Hold percentage	24.51%
Bow Out refunds paid	18,179,000 chips
Table-days equivalent	3,472.2 standard 576-round days
Raw sample exported	5,000 rounds in spitshine_audit_2m_sample.csv

Seat	Avg dealt	Avg final	Delta	AA dealt	AA final	Win rate	Total points (2M rounds)
YOU	238.3	230.1	-8.2	57,715	61,050	17.76%	482,538,440
ALICE	238.2	237.2	-1.0	57,634	70,150	19.26%	495,242,140
BOB	238.3	239.1	+0.9	57,864	71,737	19.79%	498,677,050
CAROL	238.0	245.8	+7.8	57,197	78,321	21.66%	510,213,430
DEALER	238.4	267.6	+29.2	58,193	76,198	24.66%	554,204,580

All seats are dealt identically (avg ~238 points); the dealer's two-card Shine is visible as the +29-point final-score lift. The human seat finishes lower (-8) because the default policy Bows Out weak hands, which protects bankroll but lowers the average hand played.

Hand-score tiers - dealt vs final, 2,000,000 rounds (all seats)

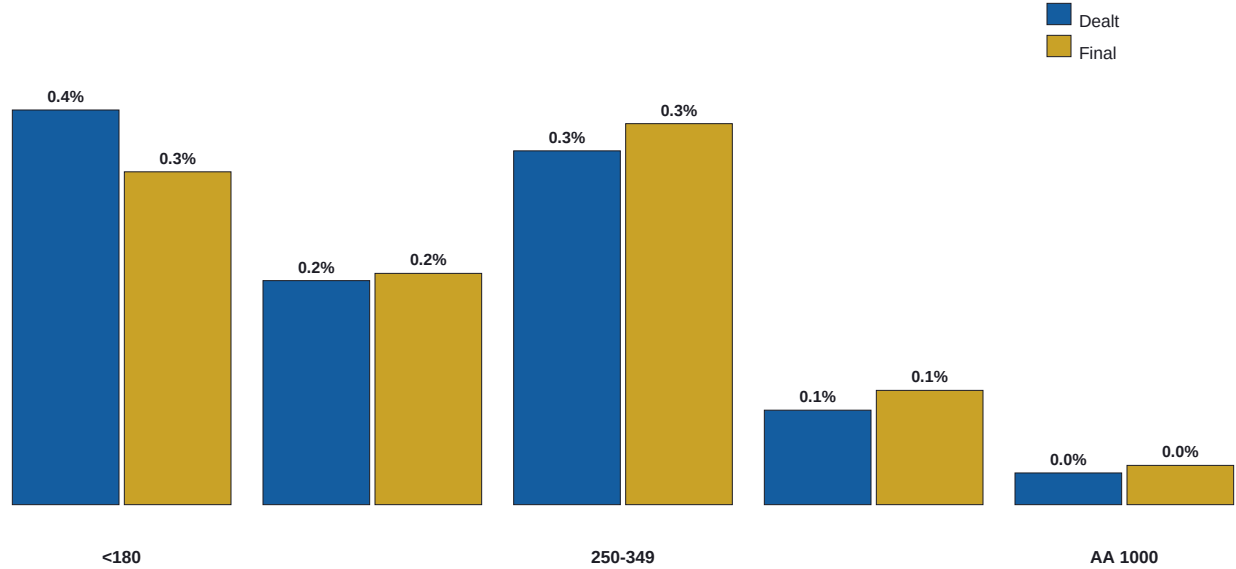
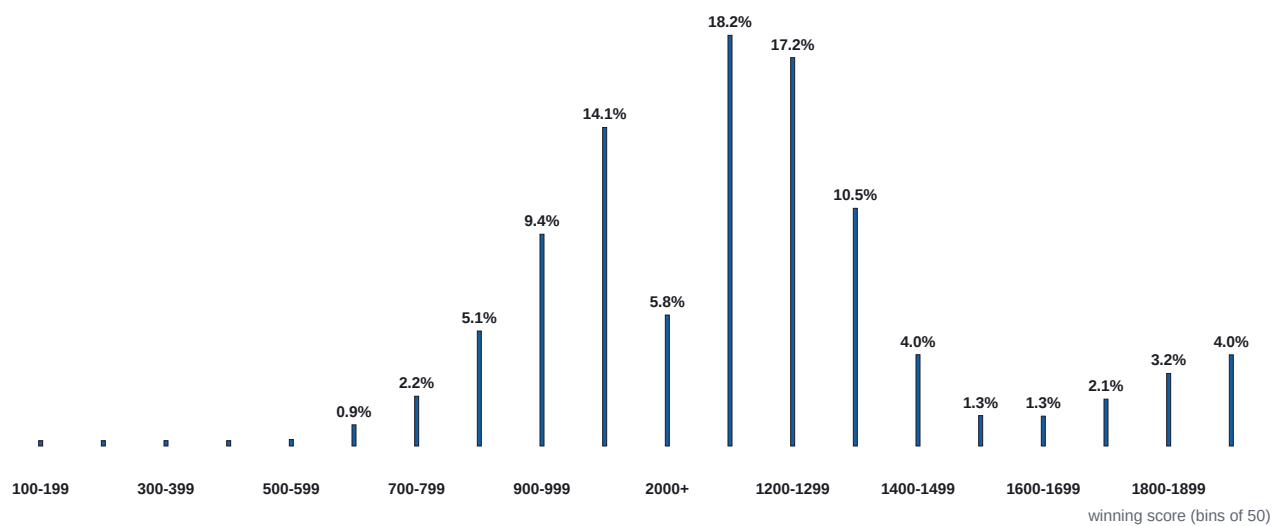


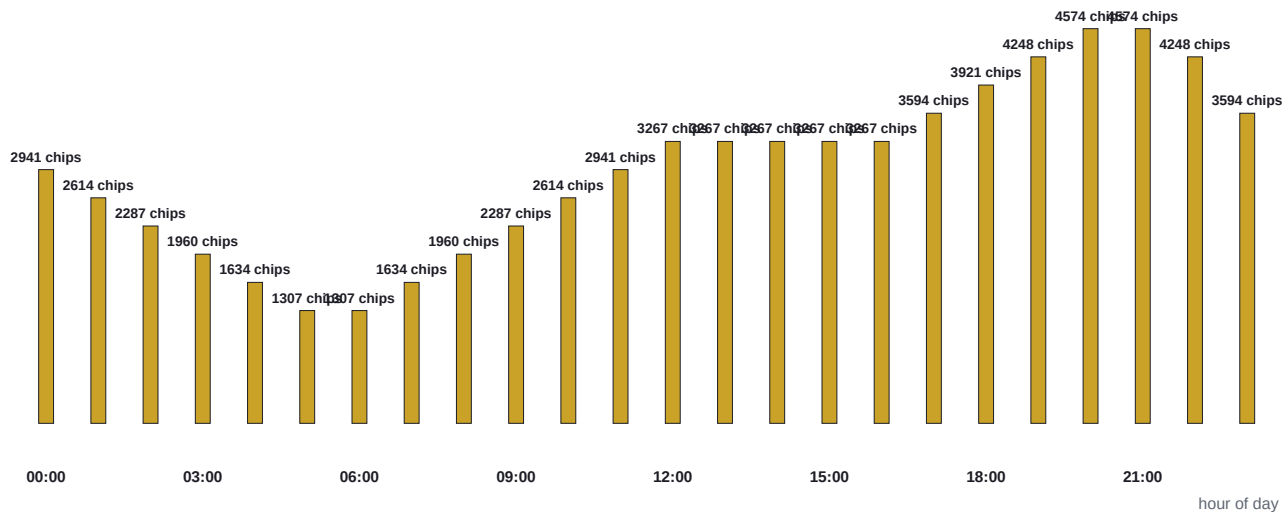
Table point total per round (sum of active seats' final scores) - 2,000,000 rounds



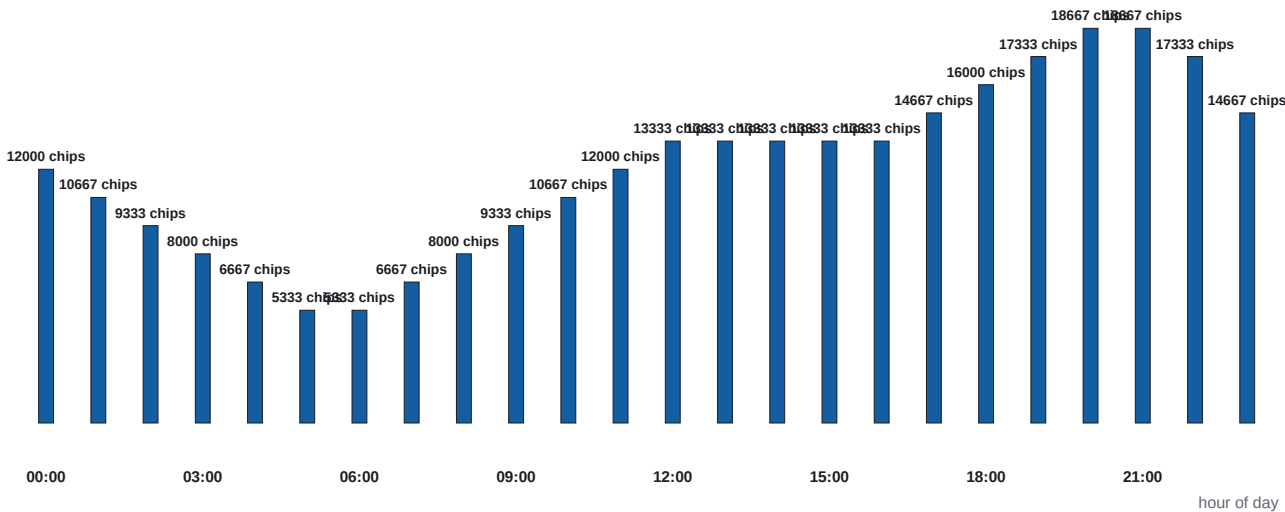
24-hour distribution

The 2,000,000-round audit is equivalent to 3,472.2 standard table-days (576 rounds/day at 24 rounds/hour). The chart below distributes one representative day across 24 hours using a typical casino traffic curve - quiet from 4am to 7am, peaking 8pm to 11pm - and the audit's verified per-round rates.

House win per hour - standard day, 100-chip ante



Handle per hour - standard day, 100-chip ante



Hour	Share	Rounds	Hands	Handle	House win
00:00	4.2%	24	120	12,000	2,941
01:00	3.7%	21	107	10,667	2,614
02:00	3.2%	19	93	9,333	2,287
03:00	2.8%	16	80	8,000	1,960

Hour	Share	Rounds	Hands	Handle	House win
04:00	2.3%	13	67	6,667	1,634
05:00	1.9%	11	53	5,333	1,307
06:00	1.9%	11	53	5,333	1,307
07:00	2.3%	13	67	6,667	1,634
08:00	2.8%	16	80	8,000	1,960
09:00	3.2%	19	93	9,333	2,287
10:00	3.7%	21	107	10,667	2,614
11:00	4.2%	24	120	12,000	2,941
12:00	4.6%	27	133	13,333	3,267
13:00	4.6%	27	133	13,333	3,267
14:00	4.6%	27	133	13,333	3,267
15:00	4.6%	27	133	13,333	3,267
16:00	4.6%	27	133	13,333	3,267
17:00	5.1%	29	147	14,667	3,594
18:00	5.6%	32	160	16,000	3,921
19:00	6.0%	35	173	17,333	4,248
20:00	6.5%	37	187	18,667	4,574
21:00	6.5%	37	187	18,667	4,574
22:00	6.0%	35	173	17,333	4,248
23:00	5.1%	29	147	14,667	3,594

Day totals: 576 rounds, 2,880 hands, 288,000 chips handle, 70,576 chips house win (24.5% hold). At \$1 chips that is about \$70.6k of table win per standard day; the peak hour (21:00-22:00) contributes about \$5.0k.

15. Integrity, Fairness & Regulatory Readiness

15.1 Randomness

The reference implementation uses a seeded Mersenne Twister for reproducibility and testing. A live casino deployment requires a certified hardware RNG (NIST SP 800-90A, FIPS 140-2/3 validated) with continuous statistical testing (NIST STS, Dieharder) and documented seeding for every shuffle.

15.2 Audit trail

Every round is recorded with card-level detail: deal, showing, Bow Out refunds, Spit swaps (who gave what to whom), Shine discards and draws, pair selections, showdown scores, tie-breaks, rake, and payouts. The log is exported as JSON and CSV from the client (AUDIT button) and is the basis for dispute resolution and regulator reporting.

15.3 Regulatory readiness checklist

Requirement	Status	Owner
Single canonical rules engine	In place (v11)	Development
Certified hardware RNG + daily statistical tests	Open	Vendor / IT
Card-level audit log with integrity (HMAC/keyed)	Log in place; keying open	Development
Shuffle / penetration / burn procedures	Documented in this guide	Operations
Independent game-math certification (e.g., GLI)	Open	Testing lab
Table limits, dispute and surveillance procedures	Open	Operations
Spit order rotation for positional fairness	Recommended - implement in floor rules	Operations
AML / KYC / responsible-gaming integration	Open	Compliance
Cooperative mode economics rebalanced or withdrawn	Open	Product

16. Edge Cases, Official Rulings & Ambiguities

This section collects every corner case identified during the audit and assigns an official ruling. Where the earlier prototypes or the design documents disagreed, the ruling below is the canonical answer; where a question cannot be settled without jurisdiction- or venue-specific policy, it is marked OPEN and must be decided before certification.

16.1 Table and seating edge cases

Situation	Ruling
Player joins during a round	Not allowed. Seats change only between rounds; the new player posts an ante and receives a fresh hand next round.
Player leaves mid-round	Not allowed for the current round; the hand plays out and the seat is removed at round end.
All players leave the table	The table closes; the dealer seat is not dealt for an empty table.
Fewer than four players	Fully supported: empty seats post no ante and take no cards; the pot and rake scale with occupied seats.
A seat posts an ante outside limits	The dealer rejects the ante; the seat must adjust to the published minimum/maximum before cards are dealt.
Mixed antes at one table	Allowed, subject to the recommended 5x spread limit (Section 14.9).
Shoe used for multiple rounds	Live floors may run a continuous shoe per Section 7; the canonical engine reshuffles every round. Per-round odds are identical either way.

16.2 Deal and card-handling edge cases

Situation	Ruling
Card exposed during the deal	Dead card: place in the discard rack and draw a replacement for that seat.
Seat dealt the wrong number of cards	Misdeal: void the round, return all antes, and redeal from the shoe (reshuffle if the cut card appears).
Duplicate cards (two decks)	Legal and expected; exact duplicates are distinct physical cards and need no special rule.
Burn card	One card is burned face-down at the start of each new shoe; it plays no part in any round.
Cut card appears mid-round	Finish the round, then reshuffle before the next deal.
Shoe exhausted mid-round	Impossible within the canonical limits (max 26 cards per round vs 104 in the shoe); if it ever occurs, void the round and reshuffle.
Player exposes a face-down card accidentally	The card stays face-up (it becomes known information) and the round continues; penalties, if any, follow house policy.

16.3 Phase-level edge cases

Phase	Situation	Ruling
2 - Showing	Player changes their mind after confirming	Confirmation is final; shown cards stay face-up for the round

Phase	Situation	Ruling
2 - Showing	Player shows all four cards	Allowed; Bow Out is locked and the seat cannot Spit or Shine (no face-down cards remain)
3 - Bow Out	Player with shown cards presses Bow Out	Rejected - showing locks Bow Out; the hand plays out
3 - Bow Out	Refund amount for a non-standard ante	25% of that seat's ante, floored to the nearest chip
4 - Spit	Actor has no face-down cards	Cannot Spit; the seat passes
4 - Spit	Target has no face-down cards	Target is not selectable; choose another seat
4 - Spit	Target already spat at this round	Not selectable - each target receives at most one Spit
4 - Spit	Human Spits at the dealer	Allowed; the dealer can receive a Spit but never initiates one
4 - Spit	Received card visibility	Private to the spitter only; never table-visible
5 - Shine	Player has no face-down cards	Cannot Shine; the seat keeps its hand
5 - Shine	Player discards without drawing	Never allowed - a replacement is always drawn from the shoe
6 - Selection	Human picks a non-optimal pair	Allowed - human choice is final; AI always optimizes
6 - Selection	Bowed-out seat selection	None required; bowed seats are excluded from showdown

16.4 Showdown and payout edge cases

Situation	Ruling	Worked example
All players Bow Out (competitive)	The dealer is the only active seat and wins	4 players bow: pot 400, rake 100, dealer receives 300 (net +200); each player -75; house +100
All players Bow Out (cooperative)	The dealer wins; the house keeps the entire pot	Pot 400, house +400; each player -100
Three-way exact tie	Pot splits three ways; remainder to house	Pot 500: payout 375 = 125 each
Two-way tie on an odd payout	Floor split; remainder to house as rounding rake	Pot 475: payout 357 = 178 + 178 + 1
Two players both hold Double Aces	Apply score, then rank, then suit; exact tie splits	A(spades)+A(hearts) beats A(diamonds)+A(clubs) on suit; identical suit-max splits
Cooperative tie (player equals dealer)	Dealer wins ties by rule	Player 300 vs dealer 300: dealer wins; house keeps pot
Payout dispute	The audit log is authoritative	Round events include every card, action, score, and payout

16.5 Digital and POC behavior

Situation	Ruling
Empty seats in the digital client	AI fills seats by default for the 4-player demo; the lobby can remove AI seats for heads-up play
Player disconnects mid-round	60-second grace period; after that the AI takes over the seat for the remainder of the round; the seat may rejoin next round
Bankroll protection on disconnect	The round settles normally; the seat's result is credited when the player reconnects
Audit access	Every round is exported to JSON/CSV via the in-game AUDIT button

16.6 Official rulings quick reference

Question	Answer
Who wins an exact tie?	The pot splits among tied seats.
Can the dealer Bow Out or Spit?	No - never.
Is the Spit-received card public?	No - private to the spitter.
Can a shown card be hidden again?	No - showing is final for the round.
What is the minimum/maximum ante?	1 chip / 500 chips.
How many players can sit?	1-4 plus the dealer; capacity 4 player seats.
When can players join or leave?	Between rounds only.
What happens if the cut card appears mid-round?	Finish the round, then reshuffle.
Who handles a payout dispute?	The floor, using the audit log.
Is the house rake always 25%?	Yes, of the pot after Bow Out refunds; rounding remainder stays with the house.

16.7 Open questions (decision required before certification)

- Jurisdictional table limits and currency conversion policy (OPEN - venue/regulator).
- Cooperative mode rebalance package approval (proposed in Section 14.5; pending formal certification).
- Side wagers and tournament variants (not part of the current rules; each requires independent math certification).
- Ante spread limit value (5x recommended; final value is a commercial decision).

17. Known Issues & Remediation

The audit identified defects across the earlier prototypes. All gameplay-critical defects are fixed in the canonical engine v11; the register below is retained for transparency and for any laboratory reviewing the development history.

ID	Severity	Issue	Status
B1	Critical	v7 GUI human Spit targeted self; v7 sim Spit was a no-op	Fixed in v11
B2	Critical	v9/v10 editions imported an unavailable copperhead module and could not launch	Superseded by v12 client
B3	High	Spat-at-once rule tracked the wrong set - multiple Spits per target possible	Fixed in v11
B4	High	Fixed Spit order created seat-position bias	Rotation recommended
B5	High	AI Bow Out missing in interactive v7/v10	Fixed in v11
B6	High	White paper odds/EV errors (Double Aces, house edge, dealer EV)	Corrected in this guide
B7	Medium	Exact ties awarded to earliest seat instead of split	Fixed in v11
B8	Medium	Cooperative all-Bow-Out crash	Fixed in v11
B9	Medium	Cooperative house edge ~46% undisclosed	Flagged for rebalance
B10	Medium	Dealer Spit inconsistent between v4 and v7/v9/v10	Canonical: dealer never Spits
B11	Low	Tie-break suit semantics inconsistent (max vs sum)	Fixed in v11
B12	Low	Integer split remainder unaccounted	Defined as rounding rake

18. Appendix A - Glossary

Term	Definition
Ante	Fixed wager placed by each seat at the start of a round (default 100).
Bow Out	Phase 3 option to forfeit and recover 25% of the ante.
Dealer	Seat 4; never Bows Out, never Spits, may Shine 0-2 cards.
Double Aces	A+A pair worth 1000 points, the highest hand.
Face-down / Face-up	Hidden versus table-visible card state.
House rake	25% of the pot taken at showdown.
Pot	Sum of antes minus Bow Out refunds.
Shine	Phase 5 discard-and-redraw; players 0-1, dealer 0-2.
Showdown	Phase 7 comparison: score, high rank, high suit, split.
Spit	Phase 4 exchange: give one face-down card, take one; received card private to the spitter.
Shoe	The 104-card dealing container.
Split	Exact-tie resolution that divides the pot equally.

19. Appendix B - Worked Example: One Full Round

The round below is a fully scripted competitive round that exercises every mechanic: Showing, Bow Out, Spit, Shine, tie-breaking, rake, and payouts. Cards are fixed for illustration; in live play every card is dealt from the shoe.

B.1 Initial deal

Seat	Cards dealt	Best pair	Tier
YOU	K(spades) Q(diamonds) 7(clubs) 3(hearts)	K+Q = 350	Strong
ALICE	A(hearts) 9(spades) 8(diamonds) 2(clubs)	A+9 = 190	Marginal
BOB	J(hearts) 7(diamonds) 5(clubs) 4(spades)	J+7 = 170	Weak
CAROL	10(spades) 6(hearts) 5(diamonds) 3(clubs)	10+6 = 160	Weak
DEALER	K(clubs) 10(hearts) 8(diamonds) 3(spades)	K+10 = 300	Solid

B.2 Phase decisions

Phase	Decision	Why
2 - Showing	YOU shows K(spades); ALICE, BOB, CAROL hide; DEALER shows K(clubs)	YOU and the dealer have 350/300 hands and show to deter Spits; the marginal hands hide to keep Bow Out
3 - Bow Out	BOB Bows Out (refund 25); ALICE and CAROL stay	BOB's 170-point hand is below the fold line; ALICE and CAROL stay despite weak hands
5 - Shine	CAROL discards 2(clubs), draws 9(clubs); everyone else keeps	CAROL's hand is 180 after her Spit; discarding the 2 and drawing 9(clubs) lifts her best pair to 10+9 = 190. YOU/ALICE/dealer hold 350/190/300 and keep

B.3 Spit swaps (Phase 4)

Actor	Target	Card given	Card taken	Hand after swap
YOU	ALICE	3(hearts)	8(diamonds)	K Q 7 8 - best pair still 350
ALICE	CAROL	2(clubs)	5(diamonds)	A 9 3 5 - best pair still 190
CAROL	DEALER	3(clubs)	8(diamonds)	10 6 2 8 - best pair 180 (improved)

Each target receives exactly one Spit: ALICE from YOU, CAROL from ALICE, DEALER from CAROL. BOB is out of the round and cannot be targeted. The dealer never Spits.

B.4 Showdown and payouts

Rank	Seat	Pair	Score	Notes
1st	YOU	K + Q	350	Winner - pot 475, rake 118, payout 357, net +257

Rank	Seat	Pair	Score	Notes
2nd	DEALER	K + 10	300	Loses 100
3rd	ALICE	A + 9	190	Beats CAROL on high rank (Ace 14 vs 10)
4th	CAROL	10 + 9	190	Tie with ALICE on score, loses on rank
Out	BOB	-	-	Bow Out: net -75

Cash-flow check: +257 (YOU) - 100 (ALICE) - 100 (CAROL) - 100 (dealer seat) - 75 (BOB) + 118 (house rake) = 0. Every chip is accounted for, and the same check is performed automatically by the audit log after every round.

20. Appendix C - Data Provenance & Reproduction

Artifact	Path / command
Canonical engine	spitshine_engine.py
Audit harness	spitshine_analysis.py --rounds 20000
Raw audit data	spitshine_odds_report.json
Headless integration test	test_aaa_flow.py
Playable client + Academy	spitshine_aaa.py
Full audit report	SPIT_SHINE_AUDIT_REPORT.md (SS-AUDIT-2026-001)

All Monte Carlo figures use the fixed seed 20260804 (competitive baseline and variants) with 20,000 rounds per scenario, and the canonical threshold policies at every seat. Exact probabilities are computed by enumeration of all $C(104,4)$ hands. Chip conservation holds to machine precision in every run.

End of document. SS-GUIDE-2026-001 v1.0 - approved for distribution to casino business development and regulatory review.